

5. Implement a Decision Tree classifier using the Iris or any relevant dataset.

CODE A:

```
#7.5.A
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

print("Libraries Loaded.")

# Load Dataset
df = pd.read_csv("IRIS DATASET.csv")
print("Dataset Loaded.")

# Features and Target
X = df.iloc[:, :-1] # All columns except last
y = df.iloc[:, -1]  # Last column (Species)

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1
)

# Create Model
model = DecisionTreeClassifier()

# Train Model
model.fit(X_train, y_train)

# Display Accuracy
print("Training Accuracy:", model.score(X_train, y_train) * 100)
print("Testing Accuracy:", model.score(X_test, y_test) * 100)

# Prediction
prediction = model.predict([X_test.iloc[0]])
print("Predicted Class:", prediction[0])
```

OUTPUT:

```
... Libraries Loaded.

Dataset Loaded.

Training Accuracy: 100.0

Testing Accuracy: 95.55555555555556

Predicted Class: setosa

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warnings.warn(
```

CODE B:

```

# 7.5.B
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor

print("Libraries Loaded.")

# Load Dataset
df = pd.read_csv("housing.csv")
print("Dataset Loaded.")

# Remove Missing Values
df = df.dropna()

# Convert categorical column into numeric
df = pd.get_dummies(df, columns=["ocean_proximity"], drop_first=True)

# Features and Target
X = df.drop("median_house_value", axis=1)
y = df["median_house_value"]

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=1
)

# Create Model
model = DecisionTreeRegressor(random_state=1)

# Train Model
model.fit(X_train, y_train)

# Display Scores
print("Training Score:", model.score(X_train, y_train))
print("Testing Score:", model.score(X_test, y_test))

# Prediction
prediction = model.predict([X_test.iloc[0]])
print("Predicted House Value:", prediction[0])

```

OUTPUT:

```

... Libraries Loaded.

Dataset Loaded.

Training Score: 1.0

Testing Score: 0.6551869659848253

Predicted House Value: 500001.0

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warnings.warn(

```